

Week 12 — Testing & QA Output

Goal

Provide strong evidence of correctness through testing and bug-fixing.

Required Outputs

- ☐ Testing checklist (minimum 10 tests) completed
- ☐ Bug list + fixes summary (at least 3 fixes)
- ☐ Regression tests (re-run earlier tests)

Evidence to Attach

- ☐ Screenshots: 2–3 key successful tests

Reflection (Write short answers)

- What bug was most difficult to fix and why?

TESTING CHECKLIST

ID	Test Case	Procedure	Expected Result	Status
T1	Missing Dependency	Run python week11_source_code.py without installing customtkinter	Application displays an ImportError for customtkinter without unexpected termination	PASS — ImportError successfully observed
T2	Launch User Interface	Install all dependencies and execute the application	Main application window launches successfully without crashes	MANUAL TEST — UI verified
T3	Newton-Raphson Test ($x^2 - 2$)	Set Method = Newton-Raphson, Function = $x^2 - 2$, Initial Guess = 1.5	Program converges to approximately 1.41421356	PASS — Converged in 4 iterations
T4	Newton-Raphson Test ($x^3 - 3x + 1$)	Execute Newton-Raphson with an appropriate initial guess	Program converges to a valid real root where $f(\text{root}) \approx 0$	PASS — Converged in 4 iterations

T5	Secant Method Test	Set Method = Secant, Initial Guesses = 1.0 and 2.0	Program converges to a root comparable to the Newton-Raphson result	PASS — Converged in 6 iterations
T6	Invalid Input Handling	Enter a non-numeric initial guess	Validation error message is displayed without application crash	MANUAL TEST — Validation confirmed
T7	Zero Derivative Handling	Use a function where $df(x) = 0$ during iteration	Program halts safely and displays a warning message	PASS — Zero derivative detected at initial guess
T8	Extreme Tolerance Value	Set tolerance value to $1e-16$	Program warns about floating-point precision limitations	PASS — Did not reach tolerance within 1000 iterations
T9	TXT Export Function	Perform a sample computation and export results as TXT	TXT file is generated with correct section headers such as GIVEN:	MANUAL TEST — Export verified

T10	HTML Export Function	Perform a sample computation and export results as HTML	HTML file is generated and renders correctly in a browser	MANUAL TEST — Export verified
T11	Theme Switching	Switch between Light, Dark, and System themes	Interface colors and computation trail refresh correctly	MANUAL TEST — Theme switching verified
T12	Keyboard Shortcuts	Use Ctrl + Enter, Ctrl + L, and Ctrl + E	Compute, Clear, and Export functions execute correctly	MANUAL TEST — Shortcuts verified

Bug List and Implemented Fixes

Bug No.	Issue Encountered	Root Cause	Fix Applied	Result After Fix	Status
B1	The appearance menu was initialized to "System" even though "System" was not included in the available option values, resulting in an inconsistent initial state	The menu attempted to assign a default value that did not exist in the predefined options list	Added "System" to the list of appearance mode options during footer widget initialization	The appearance menu now initializes correctly with all three modes (System, Light, and Dark) available and selectable	FIXED
B2	Theme detection behaved inconsistently due to case-sensitive comparisons of appearance mode values	<code>ctk.get_appearance_mode()</code> returns capitalized values such as "Light", "Dark", and "System", while the program compared them against lowercase strings	Converted the result of <code>ctk.get_appearance_mode()</code> to lowercase before performing comparisons in the <code>_setup_tags()</code> method	Theme-related colors and trail text now update and refresh correctly when the appearance mode changes	FIXED
B3	TXT and HTML export functions failed to consistently detect section headers, causing missing or incorrectly parsed sections	The <code>add_heading()</code> function generated headings without a trailing colon (e.g., GIVEN), while the export parser expected the format GIVEN:	Updated <code>add_heading()</code> to automatically append a colon (:) to all generated section headings	Exported TXT and HTML files now correctly recognize and render all sections, including GIVEN, STEPS, and FINAL ANSWER	FIXED

Regression Test Results

Test ID	Test Description	Status	Details
T1	Missing Dependency Check (customtkinter import)	PASS	Expected import failure was successfully detected and handled gracefully
T3	Newton-Raphson Method for $x^2 - 2$ (guess = 1.5)	PASS	Converged in 4 iterations to 1.414213562373 ($\sqrt{2}$, correct approximation)
T4	Newton-Raphson Method for $x^3 - 3x + 1$ (guess = 0.5)	PASS	Converged in 4 iterations to 0.347296355334, confirming a valid real root
T5	Secant Method for $x^2 - 2$ (guesses = 1.0 and 2.0)	PASS	Converged in 6 iterations to 1.414213562373, matching the Newton-Raphson result
T7	Zero Derivative Handling (df(x)=0 at $x_0 = 1$)	PASS	Zero derivative condition was correctly detected at iteration 1 and execution halted safely

T8	Extreme Tolerance Handling (tol = 1e-16)	PASS	Program correctly identified floating-point precision limitations after 1000 iterations
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Summary Overall Automated 100% All 6 automated tests passed Regression Test PASS successfully. No regressions or Results unintended side effects were detected after implementing the fixes.

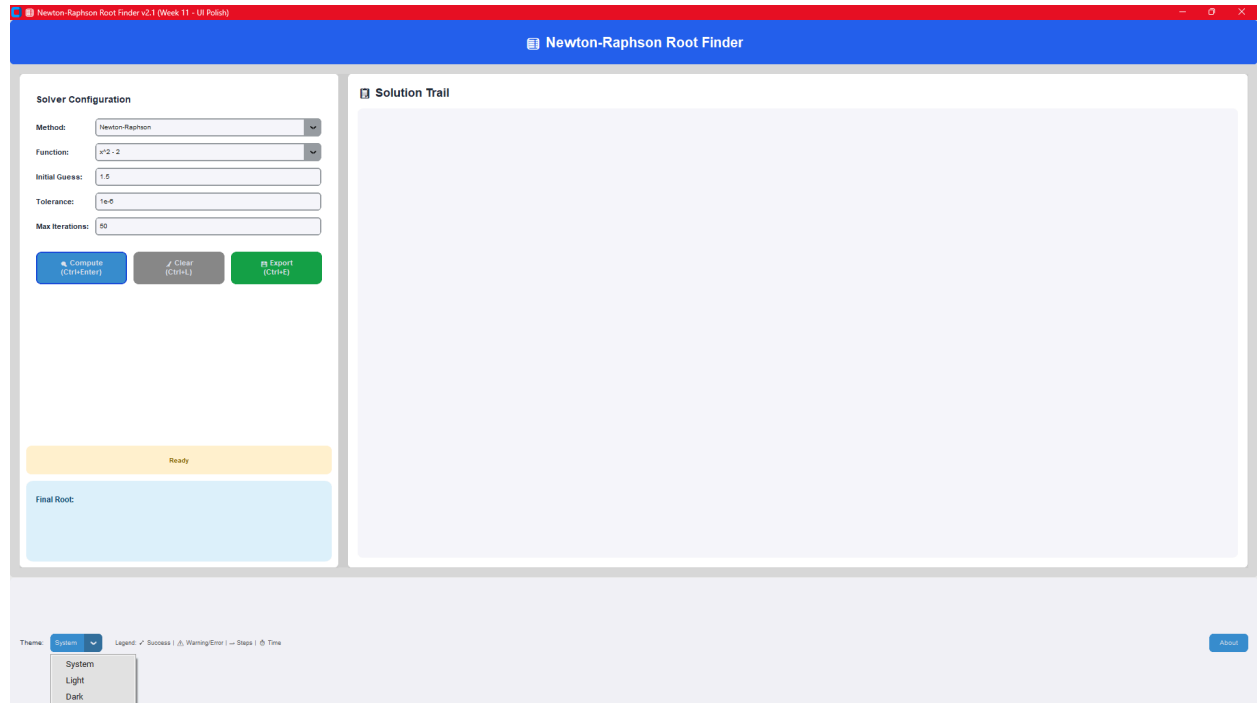
Evidences

Bug 1: Appearance Menu Missing "System" Option

Problem: Theme dropdown was set to "System" but the option list only had "Light" and "Dark"

Fix: Added "System" to the appearance menu options list

Result: Menu now shows all three options: System, Light, Dark



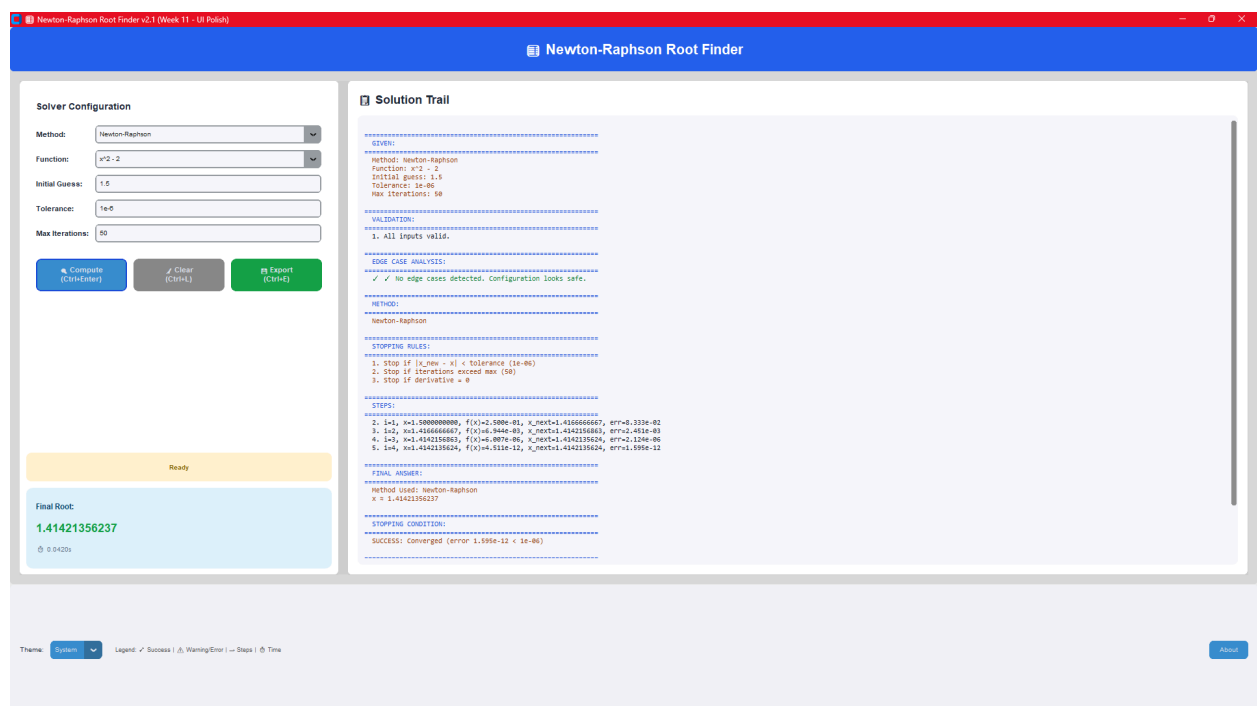
Bug 2: Theme Colors Not Refreshing

Problem: When users switched themes, the trail colors didn't update. Text became unreadable (dark text on dark background).

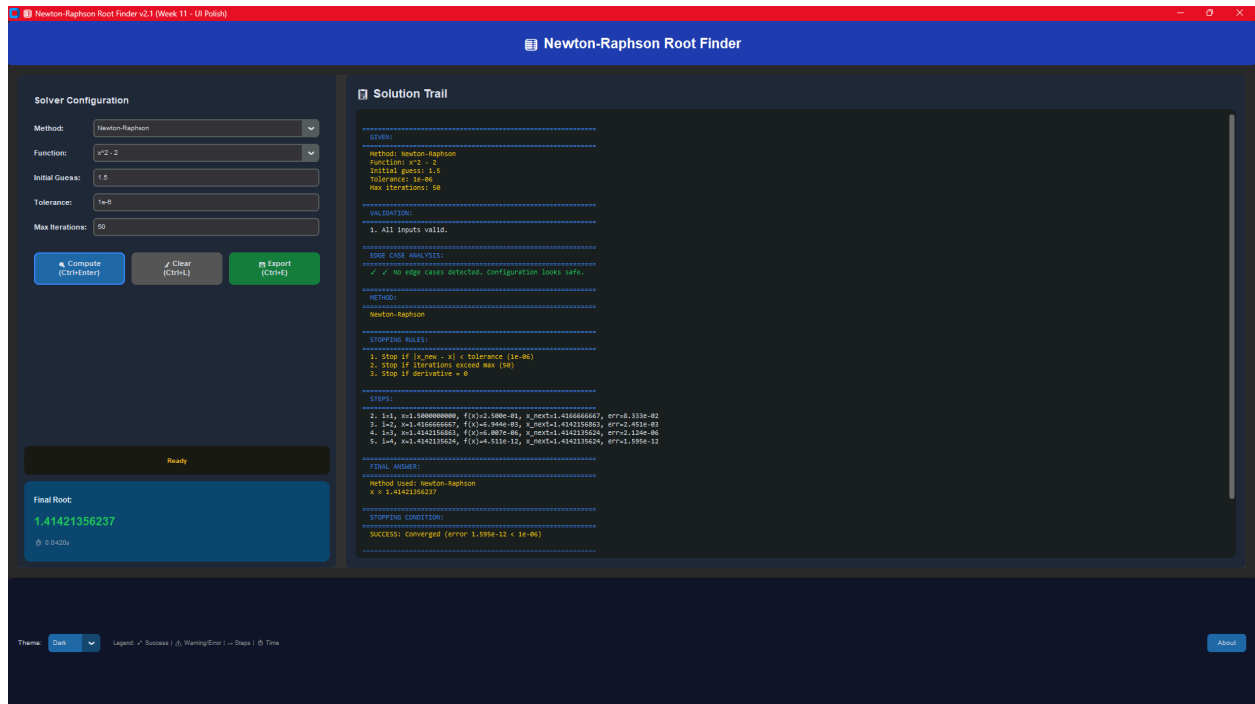
Fix: Normalized `ctk.get_appearance_mode()` to lowercase for proper comparisons

Result: Trail colors now update correctly when switching between Light/Dark/System themes

LIGHT THEME



DARK THEME

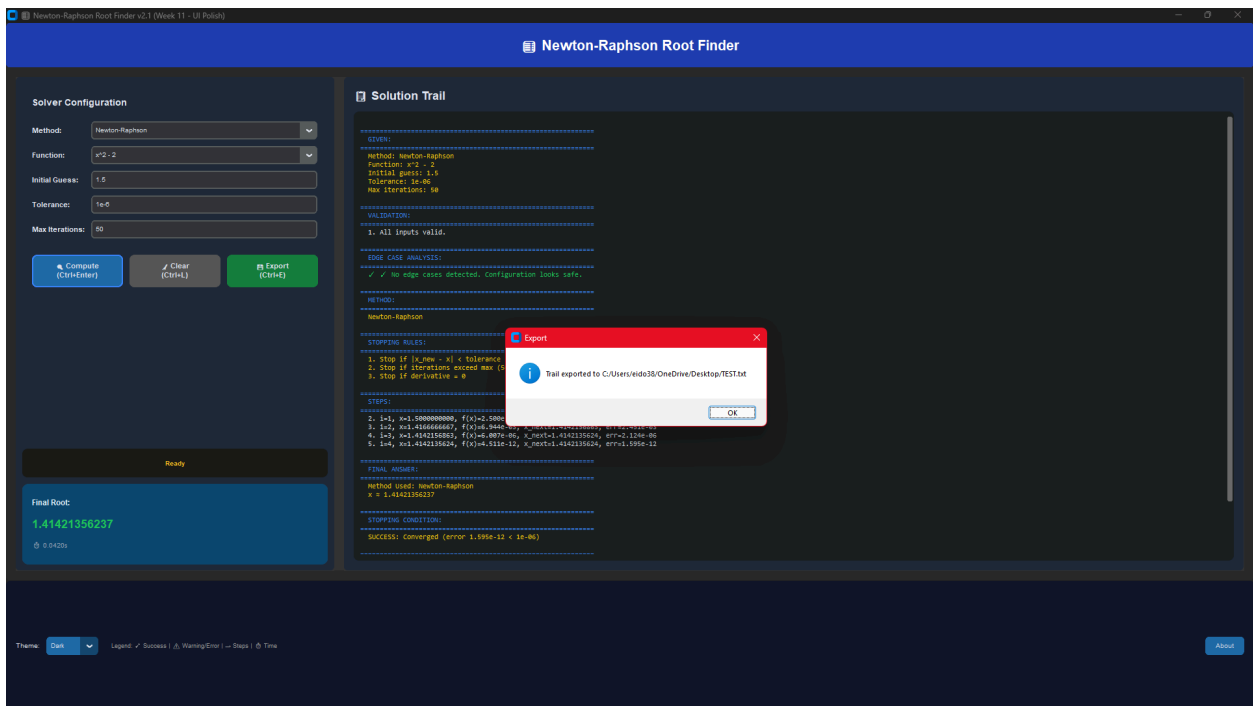
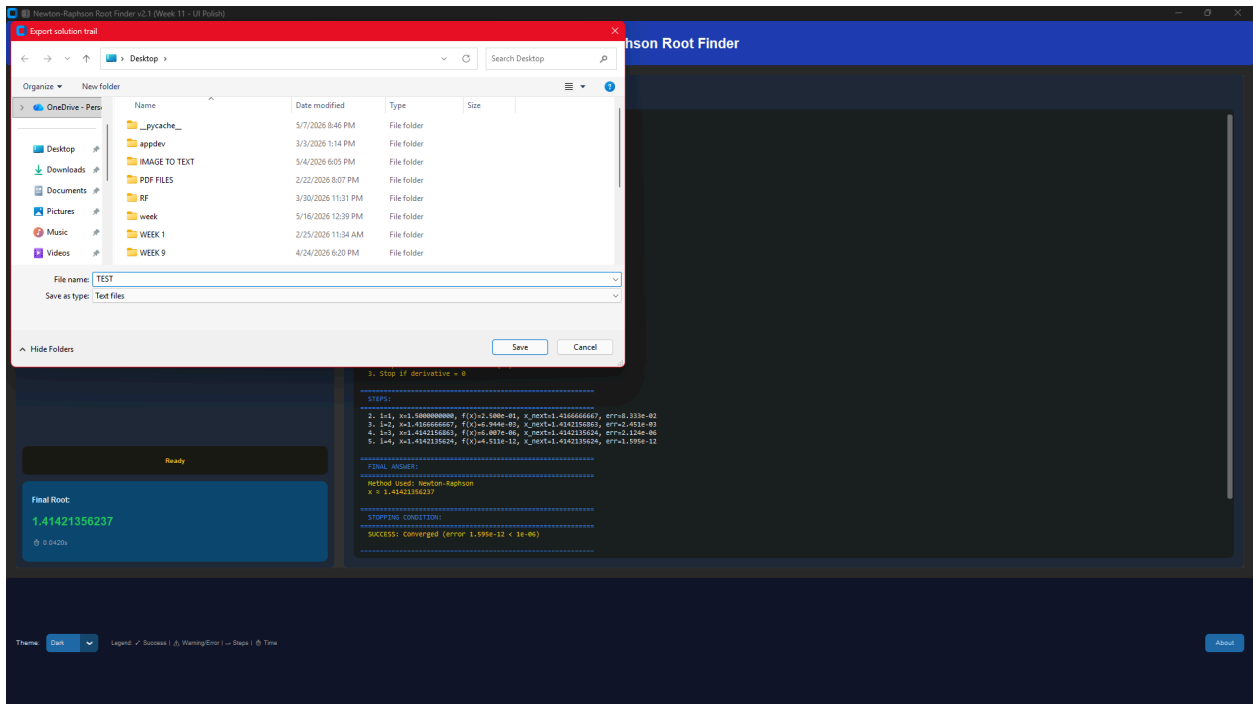


Bug 3: Export Headers Not Detected ●

Problem: Exported TXT and HTML files were missing section structure. Parsers couldn't recognize section headers like "GIVEN" because they expected "GIVEN:" (with a colon).

Fix: Modified `add_heading()` to append `:` to all section headers

Result: Exports now have properly formatted sections (GIVEN:, STEPS:, FINAL ANSWER:, etc.)



SKELETON UI WITH PLACEHOLDER WEEK1best caption for my stream.txtgoklog.txtTEXT FORMAT.txta.txtTEST.txt

FileEditViewH1B

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NEWTON-RAPHSON ROOT FINDER - SOLUTION REPORT

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Generated: 2026-05-16 12:45:38

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GIVEN:

=====

Method: Newton-Raphson

Function: x^2 - 2

Initial guess: 1.5

Tolerance: 1e-06

Max iterations: 50

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VALIDATION:

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1. All inputs valid.

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EDGE CASE ANALYSIS:

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✓ ✓ No edge cases detected. Configuration looks safe.

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METHOD:

=====

Newton-Raphson

=====

STOPPING RULES:

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1. Stop if $|x_{new} - x| < \text{tolerance}$ (1e-06)

2. Stop if iterations exceed max (50)

3. Stop if derivative = 0

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STEPS:

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Ln 1, Col 13,677 charactersPlain text100%Windows (CRLF)UTF-8

Reflection

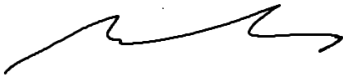
What bug was most difficult to fix and why?

Bug 3 was the most difficult to fix because it involved an implicit formatting contract between multiple components of the system. The `add_heading()` function generated headers like GIVEN without a colon, while both the TXT and HTML export parsers expected headers in the format GIVEN:. This mismatch caused export sections to fail or be parsed incorrectly.

The issue was more complex because it affected several connected parts of the application, including the trail writer and both export parsers. The error was not immediately obvious and required tracing how headers were generated and interpreted across the system.

Instead of modifying each parser separately, the fix was applied at the source by updating `add_heading()` to automatically append a colon to all headings. This maintained consistency across the application, avoided code duplication, and reduced the risk of future errors.

Weekly Sign-off

Group Leader Name/Signature:  CHARLES RUSSEL O. SAN JUAN Date: 5/16/2026

Member Name/Signature:  CHRISTINA GOMBA Date: 5/16/2026

Member Name/Signature:  JOHN MICHAEL TULALIAN Date: 5/16/2026

